D 13154 **(Pages : 2)** **Name.....**

Reg. No.....

FIRST SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2021

(CBCSS)

Microbiology

MBG 1C 02—BIOPHYSICS AND INSTRUMENTATION

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

General Instructions

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section / Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section / sub part / sub division may be ignored.*
4. *There will be an overall ceiling for each Section / Part that is equivalent to the maximum weightage of the Section / Part.*

Where ever needed answers must be supported by structural illustrations and diagrams.

Section A (Short Answer Type)

Answer any **four** *of the following*
Each question carries 2 *weightage.*

1. HPLC.
2. Peptide linkage.
3. Ultra filtration.
4. Confocal Microscopy.
5. Flame photometry.
6. Tortion angle.

(4 × 2 = 8 weightage)

Turn over

Section B (Short Essay Type Questions)

Answer any **four** *of the following.*
Each question carries 3 *weightage.*

Comment on the following :

7. Brief account on the concept of Entropy and Enthalpy.
8. Differentiate Alpha helix and Beta sheets.
9. Give details on DNA-Histone interaction.
10. Differentiate SDS and Native PAGE.
11. Dialysis and its applications in biology.
12. Give details on the principle of X-Ray diffraction technique.

(4 × 3 = 12 weightage)

Section C (Essay Type Questions)

Answer any **two** *questions.*
Each question carries 5 *weightage.*

13. Write detail on principle, design and applications of various spectroscopy techniques.
14. Give details on various physico-chemical forces seen in bio molecules.
15. Describe on protein- drug interactions studies using databases and PMF techniques.
16. Give details on various centrifugation techniques.

(2 × 5 = 10 weightage)